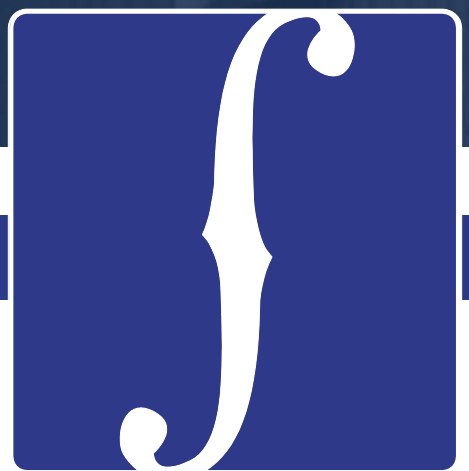
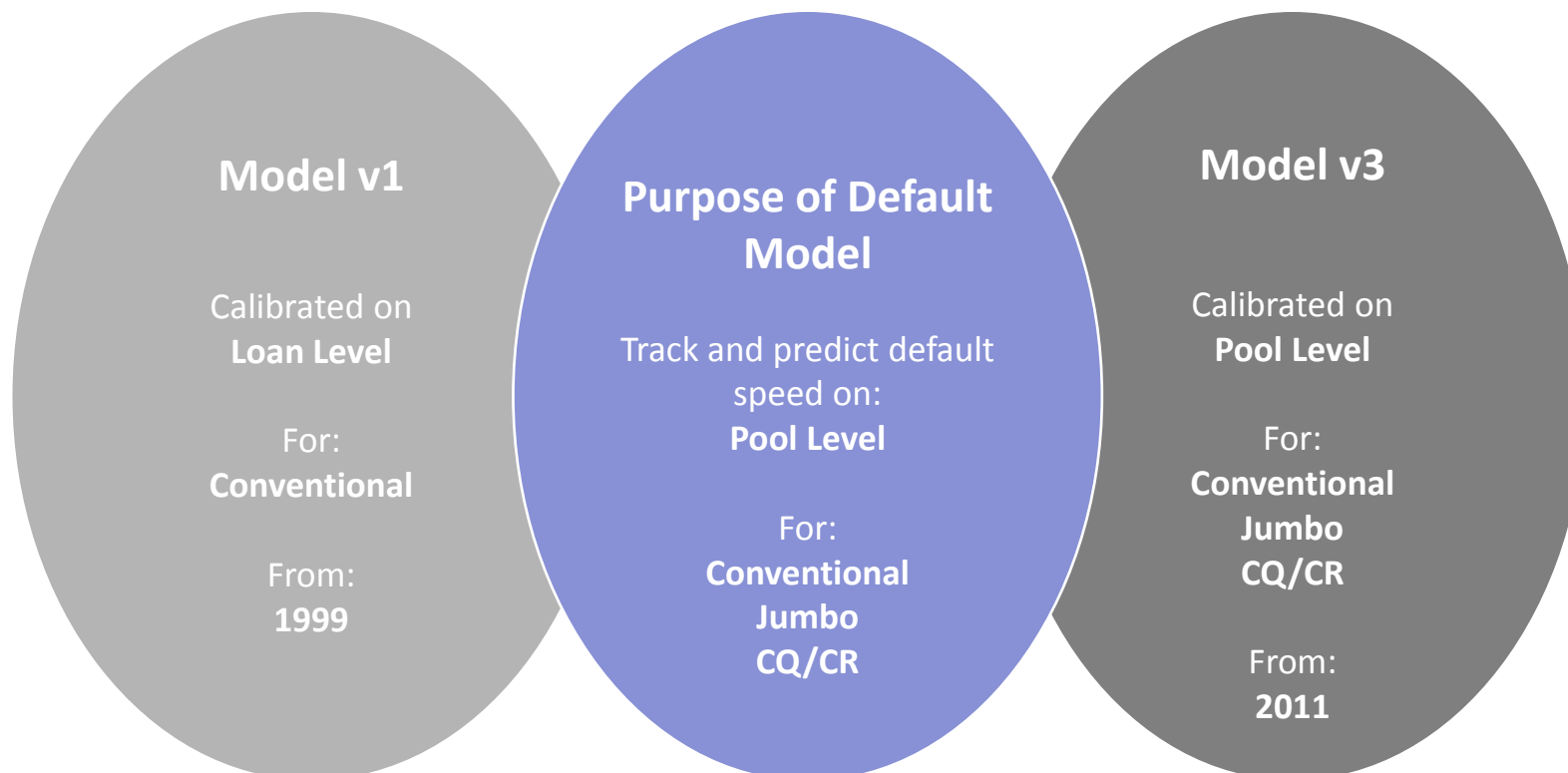




CELLO

Agency Prepayment: Default

March 2016



Longer history	Strengths	- Pool level model for pool level tracking - Calibrated on Jumbo, CQ/CR
- Using loan level model to calculate pool level default - Not calibrated to Jumbo, CQ/CR data set	Weaknesses	Limited historical data

Default model V1

2 Sub-models

Model for current loans
Model for delinquent loans

4 Parameters

CLTV
FICO
WALA
Occupancy

2 Adjustment Functions

Tweaks for CLTV, WALA
Decay function for CLTV, WALA



Changes

1. Add plausible long-range forecast : Months Since Starting Payment Status

For each starting payment status (current/delinq), we build a curve to show how default rates evolve over time.

2. Remove occupancy parameter

Based on our fits, Occupancy status is not significant for the pool level default data set.

3. Add Credit Availability parameter

When there is more credit availability (CA), it tends to be easier for borrowers to refinance instead of default. Regression results support a negative correlation between CA and the default rate.

4. Remove adjustment functions

The old model involved tweaks because we used loan level data to calibrate the model and then used these adjustment factors to fit pool level data

5. Add constraints for extrapolation

Using cap and floors for out of range extrapolation. Make sure the tail performance is reasonable.

Default model v1

2 Sub-models

Model for current loans
Model for delinquent loans

4 Parameters

CLTV
FICO
WALA
Occupancy

2 Adjustment Functions

Tweaks for CLTV, WALA
Decay function for CLTV, WALA



Default model v3

2 Sub-models

$$\text{CDR} = \text{Current\%} * \text{Model_CurrentLoans} + \text{Delinq\%} * \text{Model_DelqLoans}$$

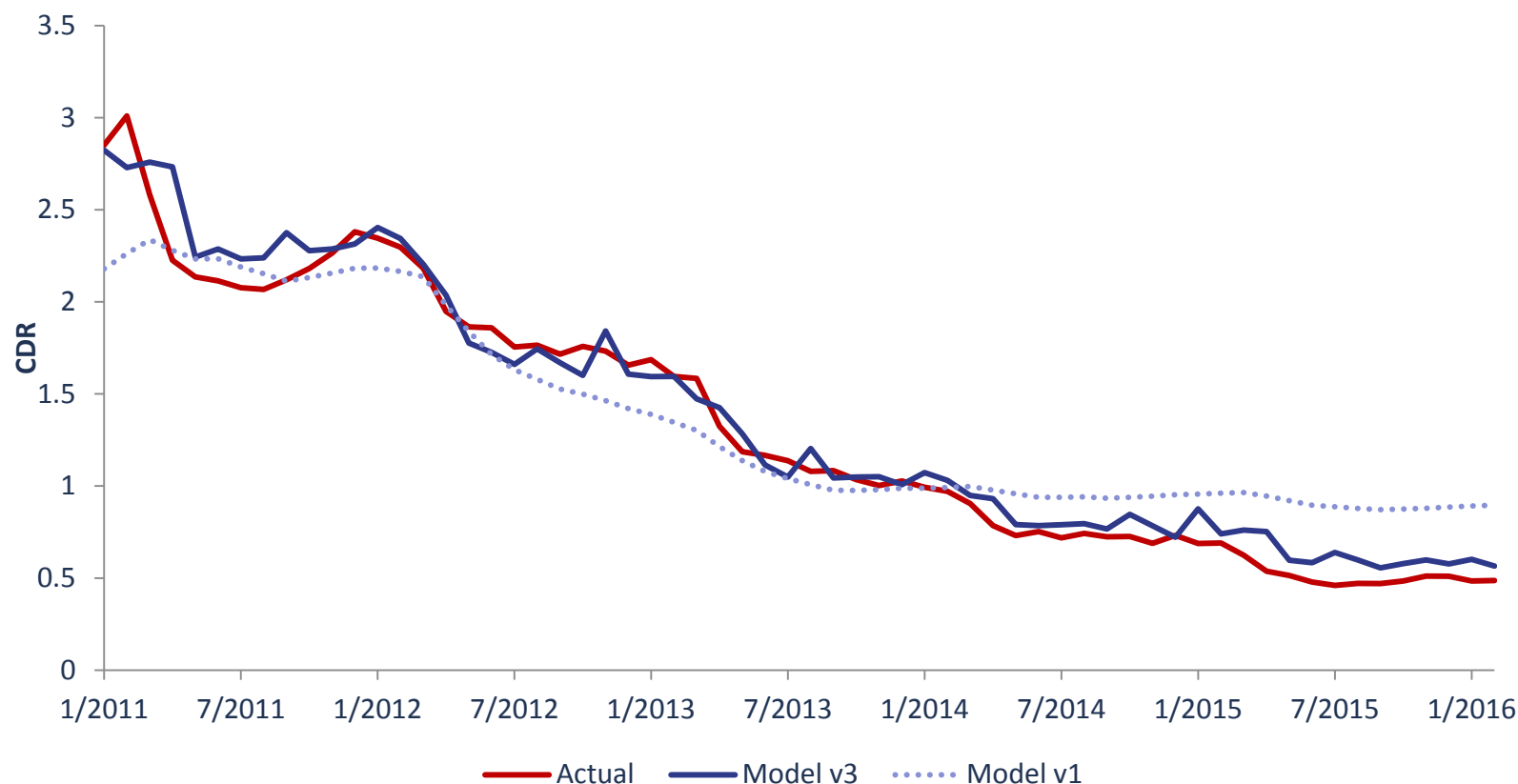
5 Parameters

MonthSinceStartingPaymentStatus
CLTV
Credit Availability
FICO
WALA

1 Constraint

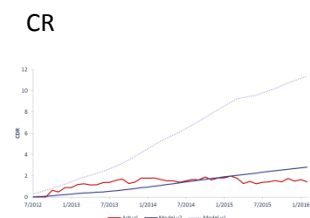
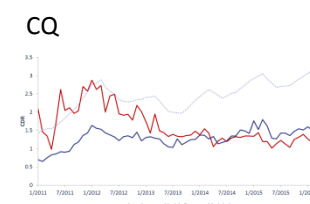
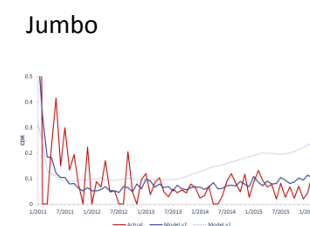
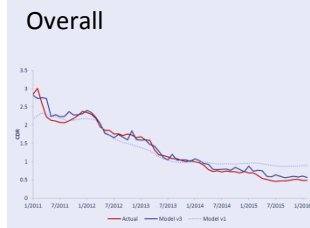
Constraints for extrapolation

Model Tracking Performance – Overall (Conventional + Jumbo + CQ/CR)

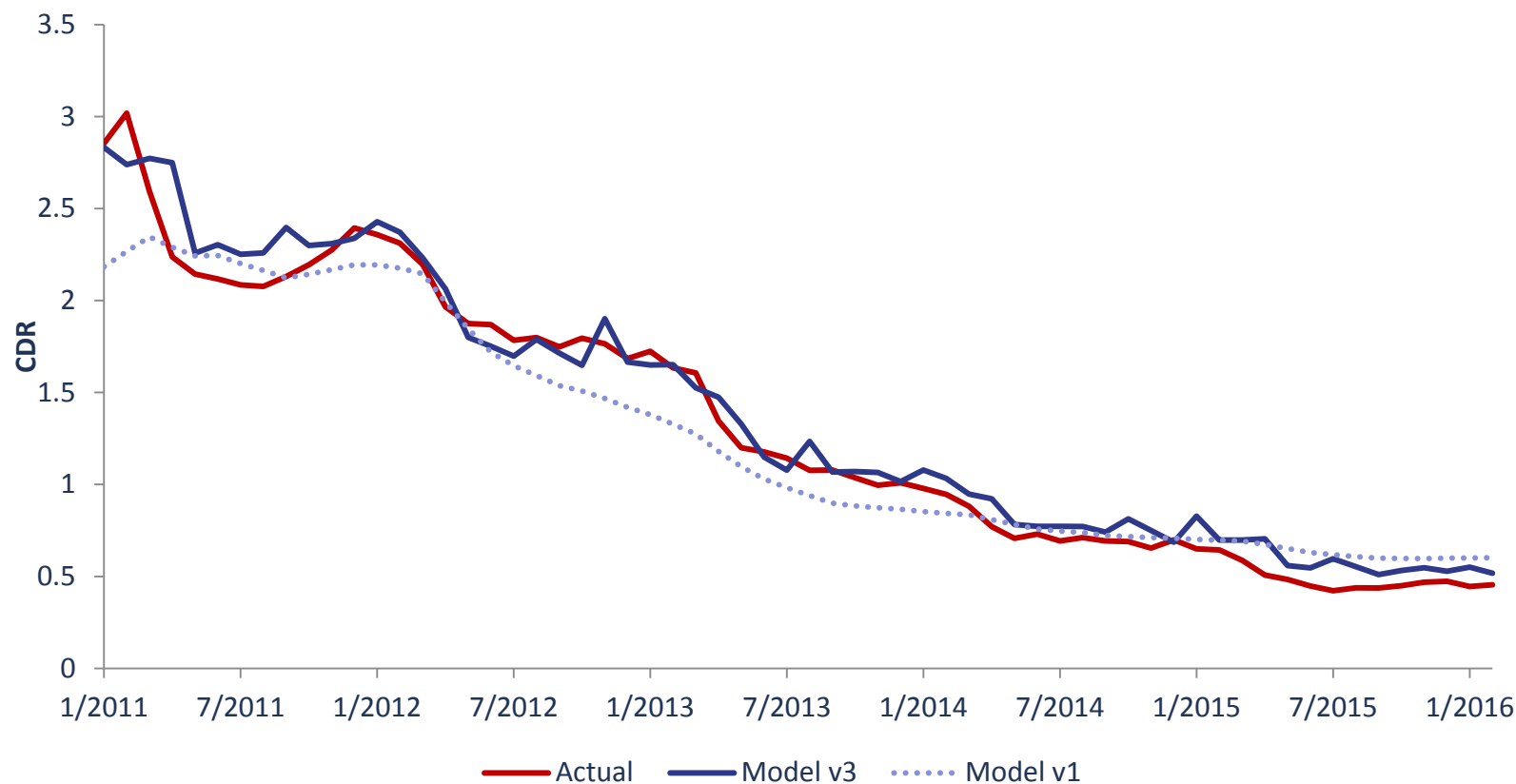


Model v3 has better performance in two areas:

- 1) For the most recent 2 years, model v1 is too fast, while v3 has better tracking
- 2) Model v3 is better at capturing the peak default rate (in pool level data) in 2011.



Model Tracking Performance – Conventional



Both of the two models have good performance for conventional (CL) pools.
The tracking error for model v3 is lower, especially for the most recent two years.

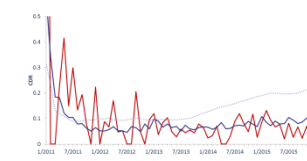
Overall



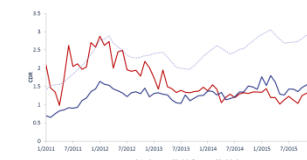
Conventional



Jumbo



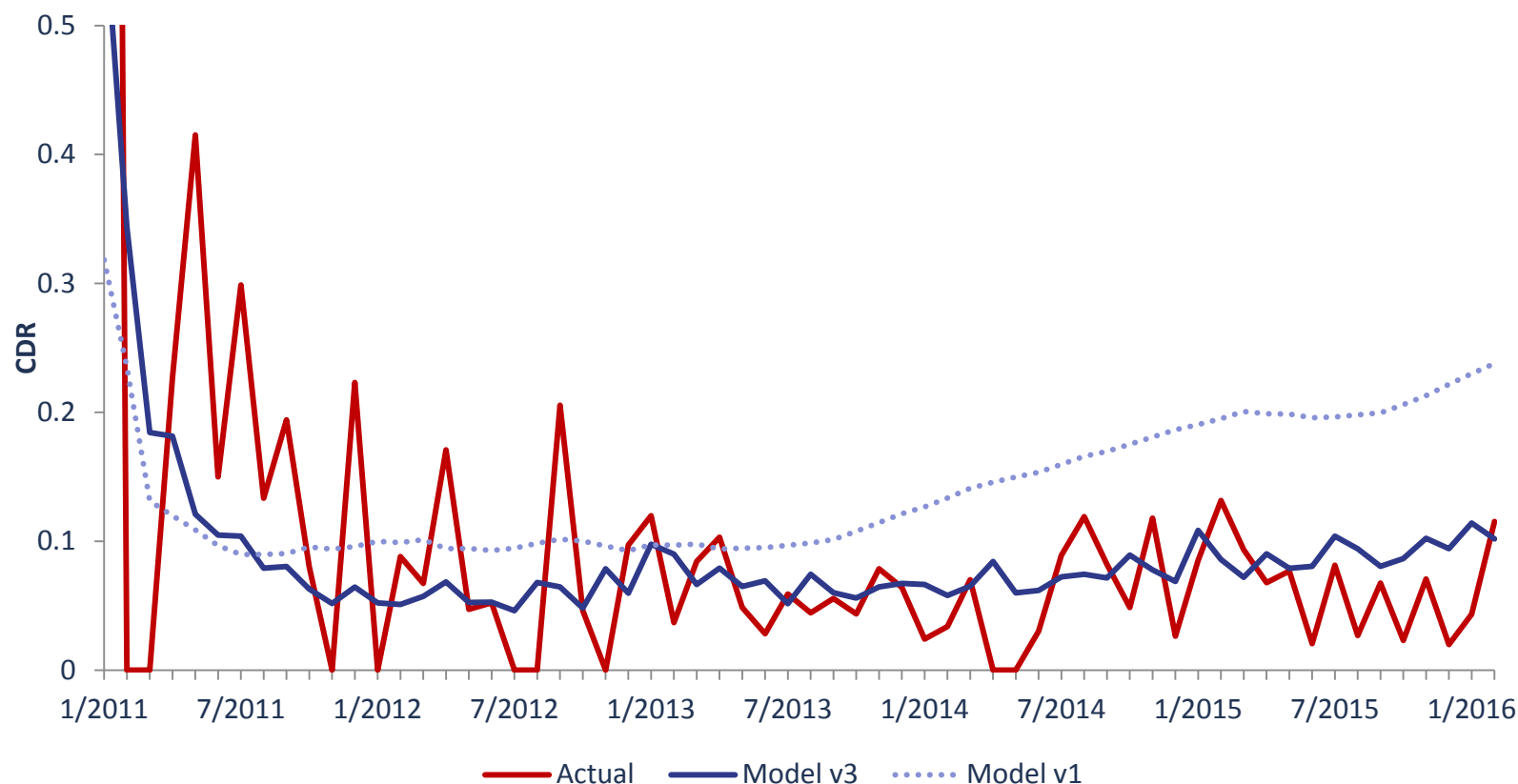
CQ



CR



Model Tracking Performance – Jumbo

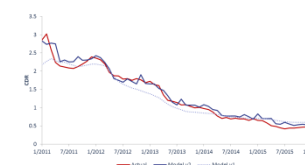


Both of the two models have a good performance for conventional Jumbo pools .
The tracking error for model v3 is lower.

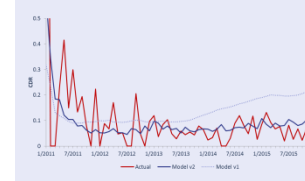
Overall



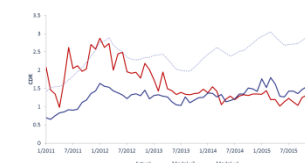
Conventional



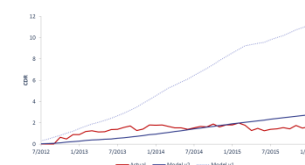
Jumbo



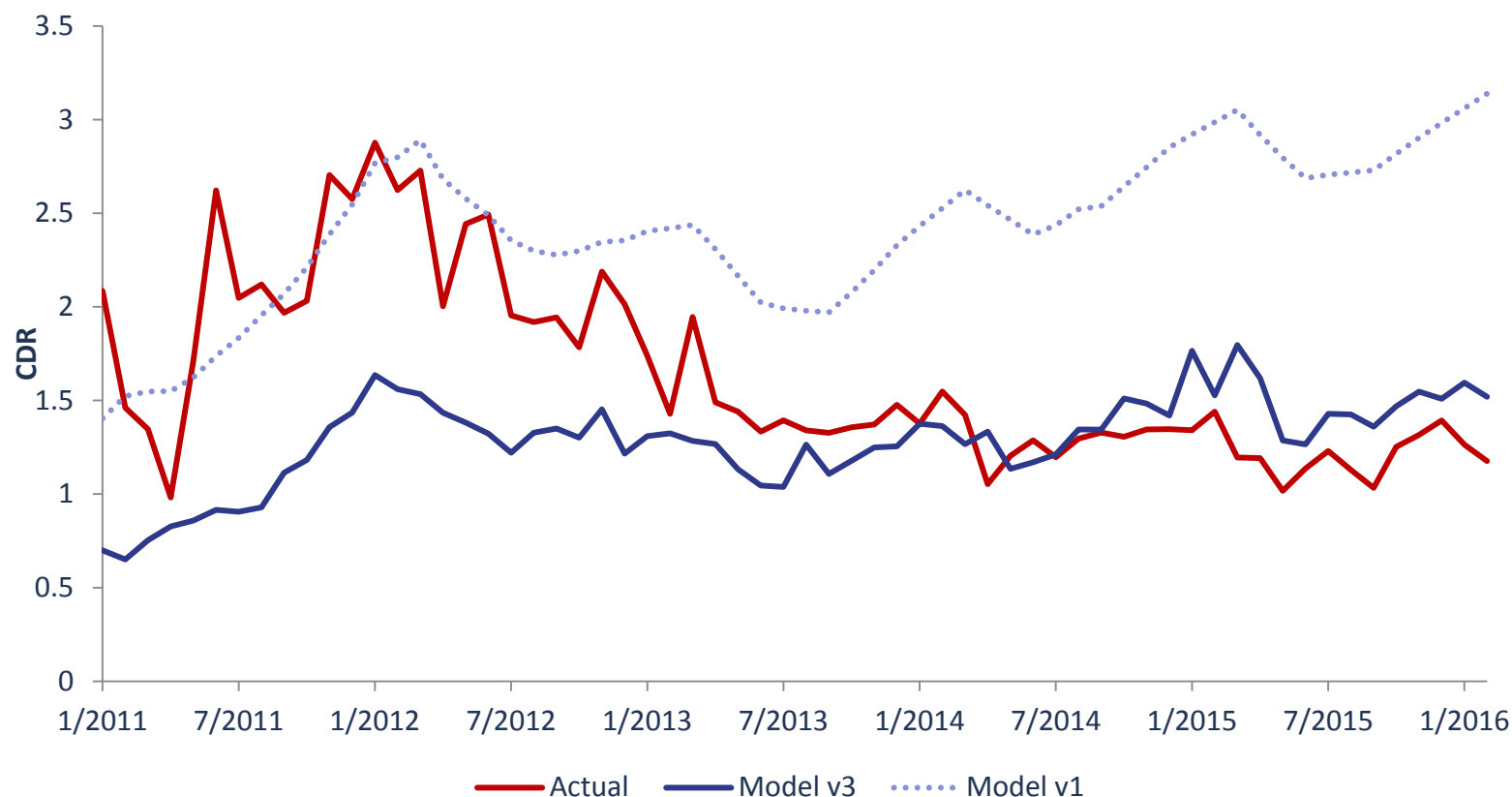
CQ



CR



Model Tracking Performance – CQ

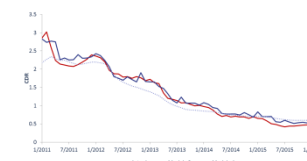


For CQ pools, both of the models are too fast after 2014, but model v3 is closer to actual prepayment speeds in that period. However, Model v1 displays better performance in 2011/2012.

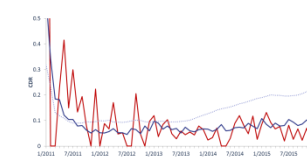
Overall



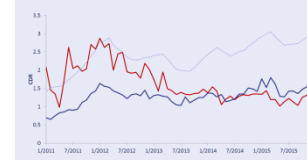
Conventional



Jumbo



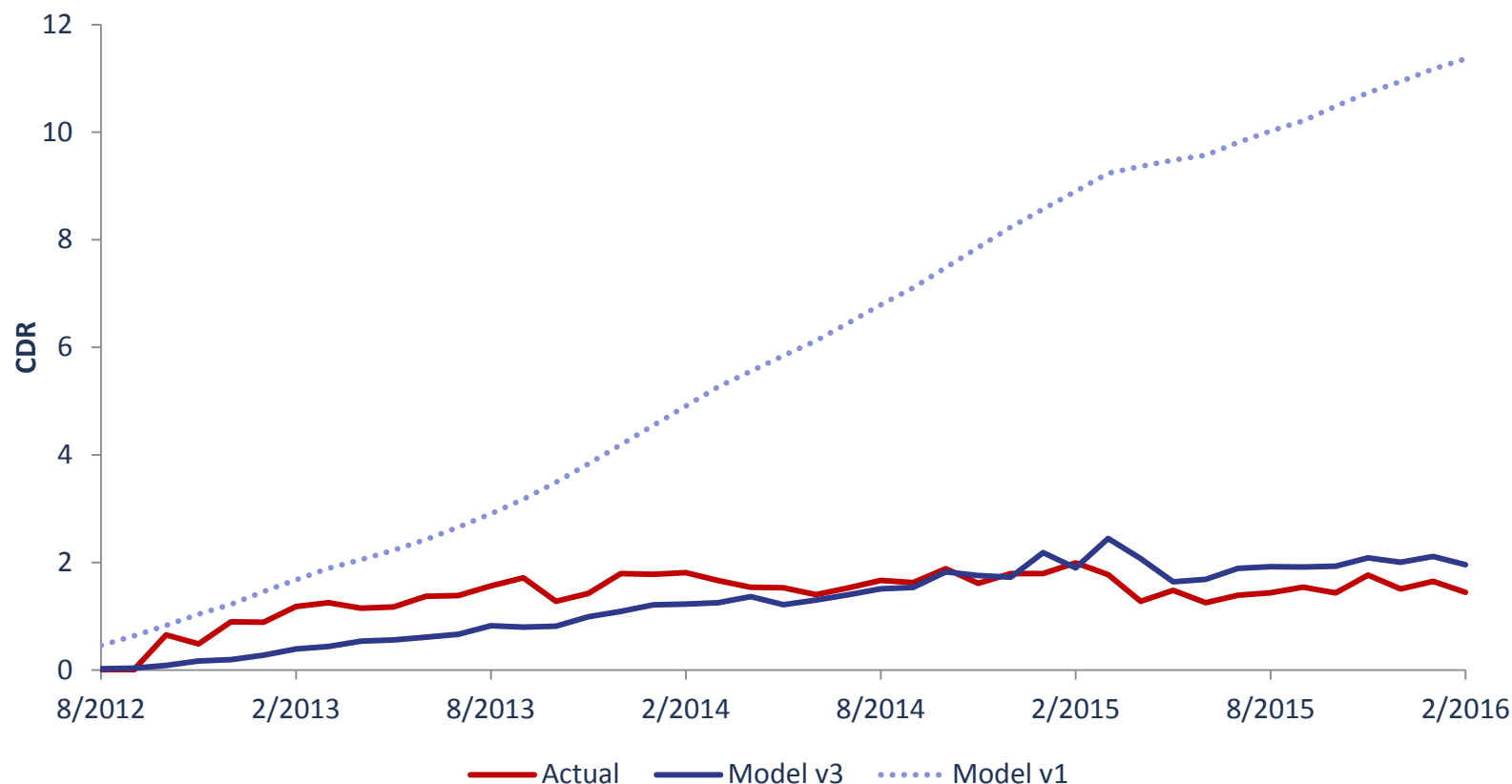
CQ



CR

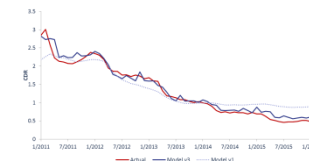


Model Tracking Performance – CR



Model v3 has a much better performance on CR pools compared with v1. Some of the underperformance by v1 is an artefact of the fact that significant extrapolation was necessary given that v1 was only fitted on CL loans.

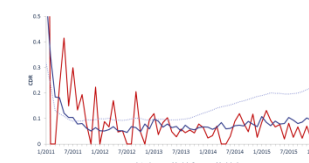
Overall



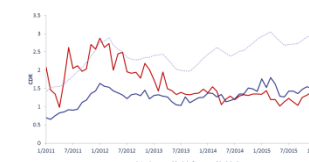
Conventional



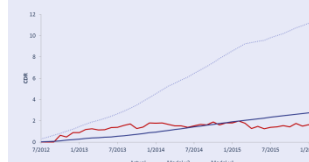
Jumbo

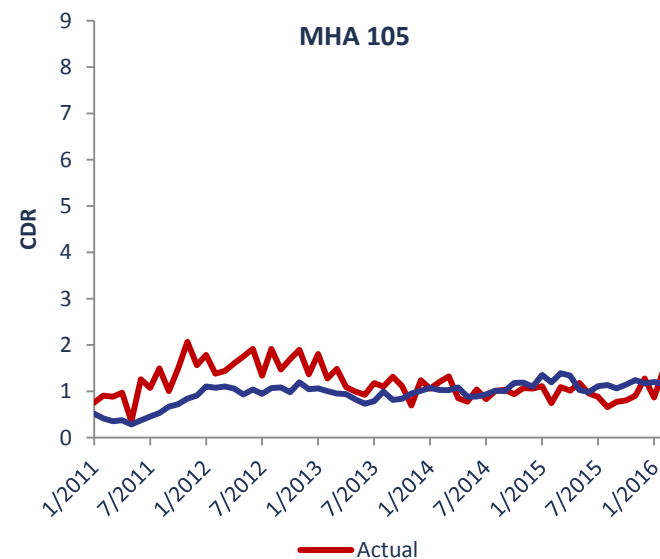
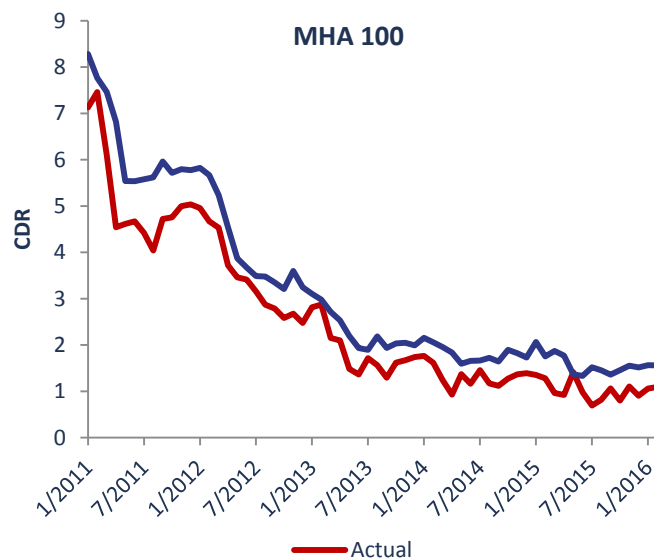
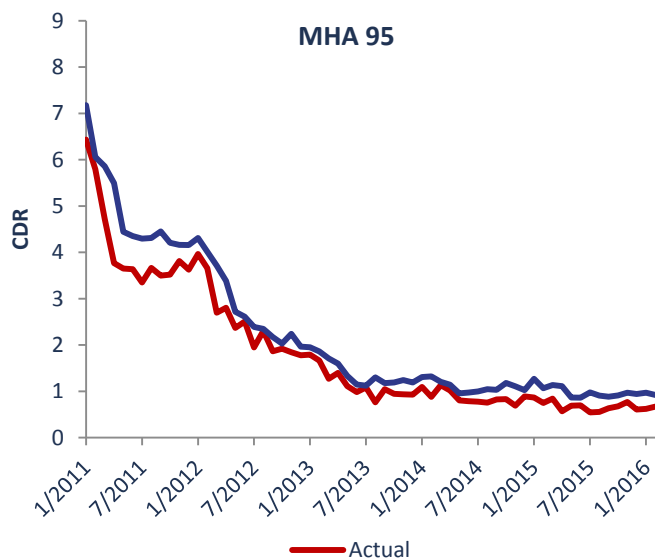
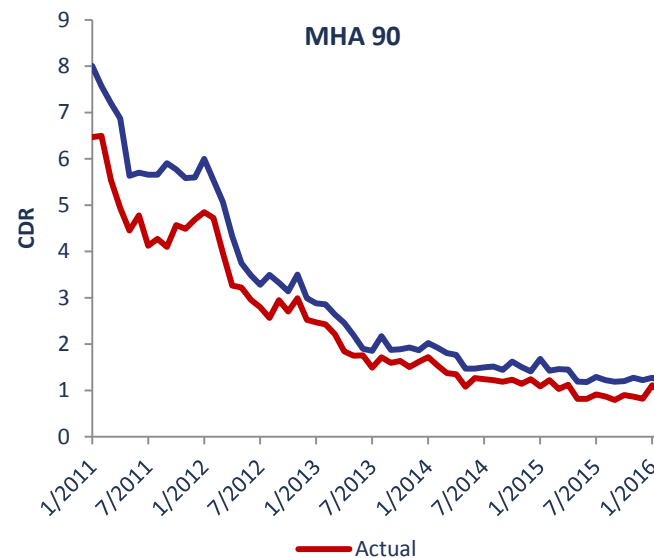
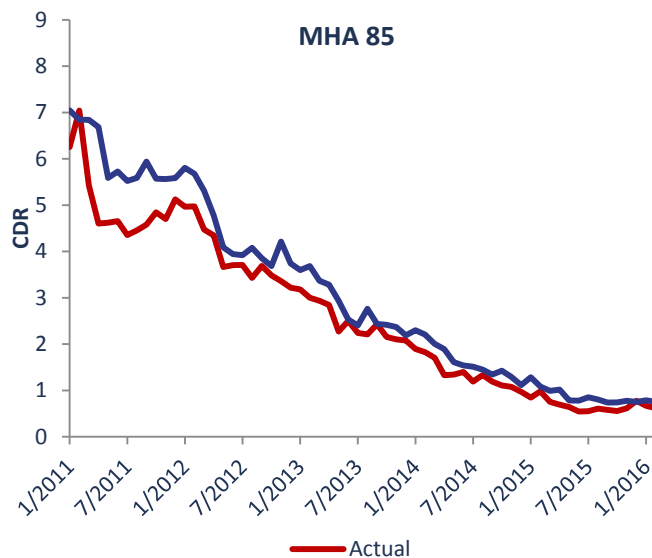


CQ

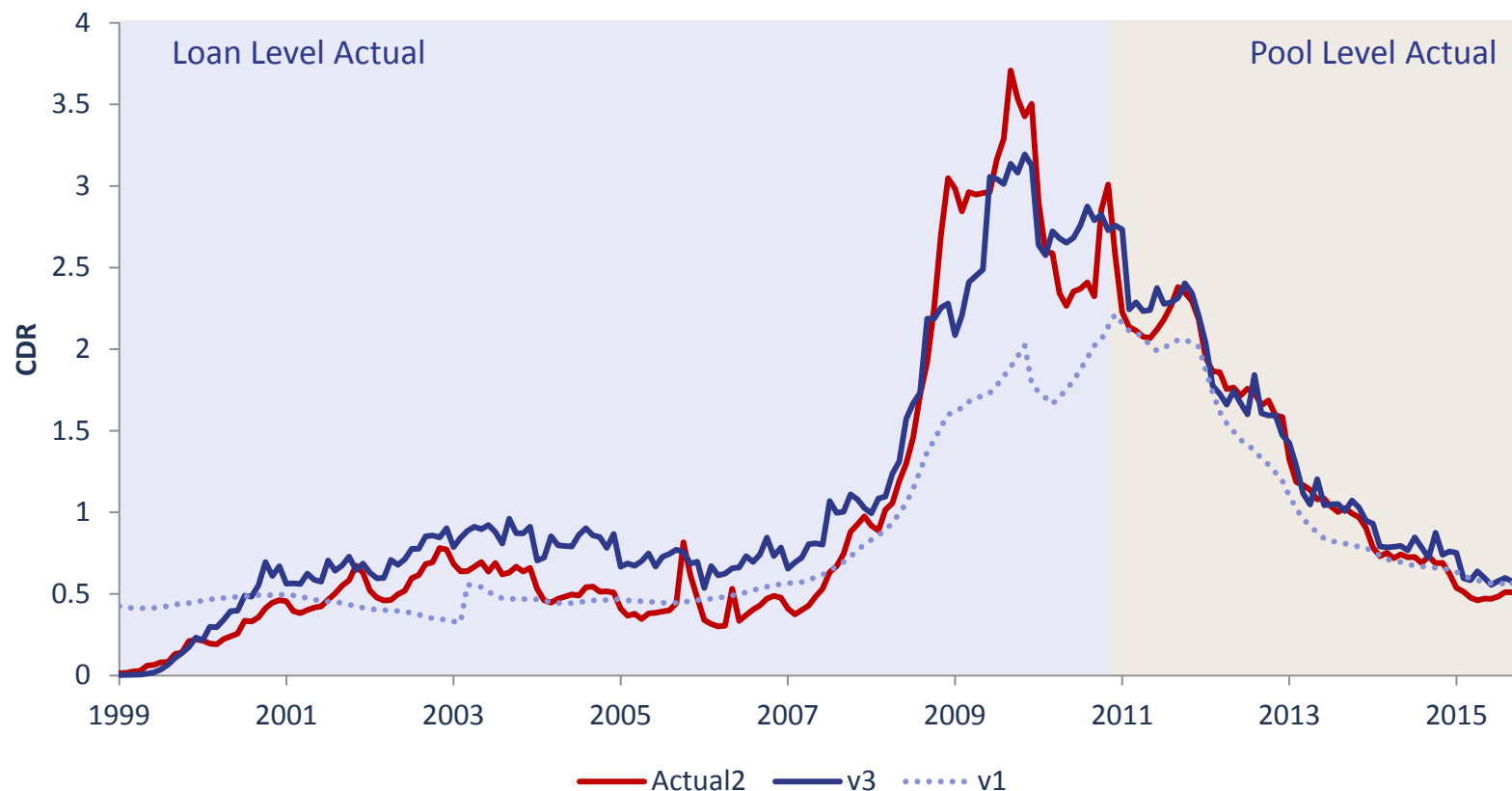


CR



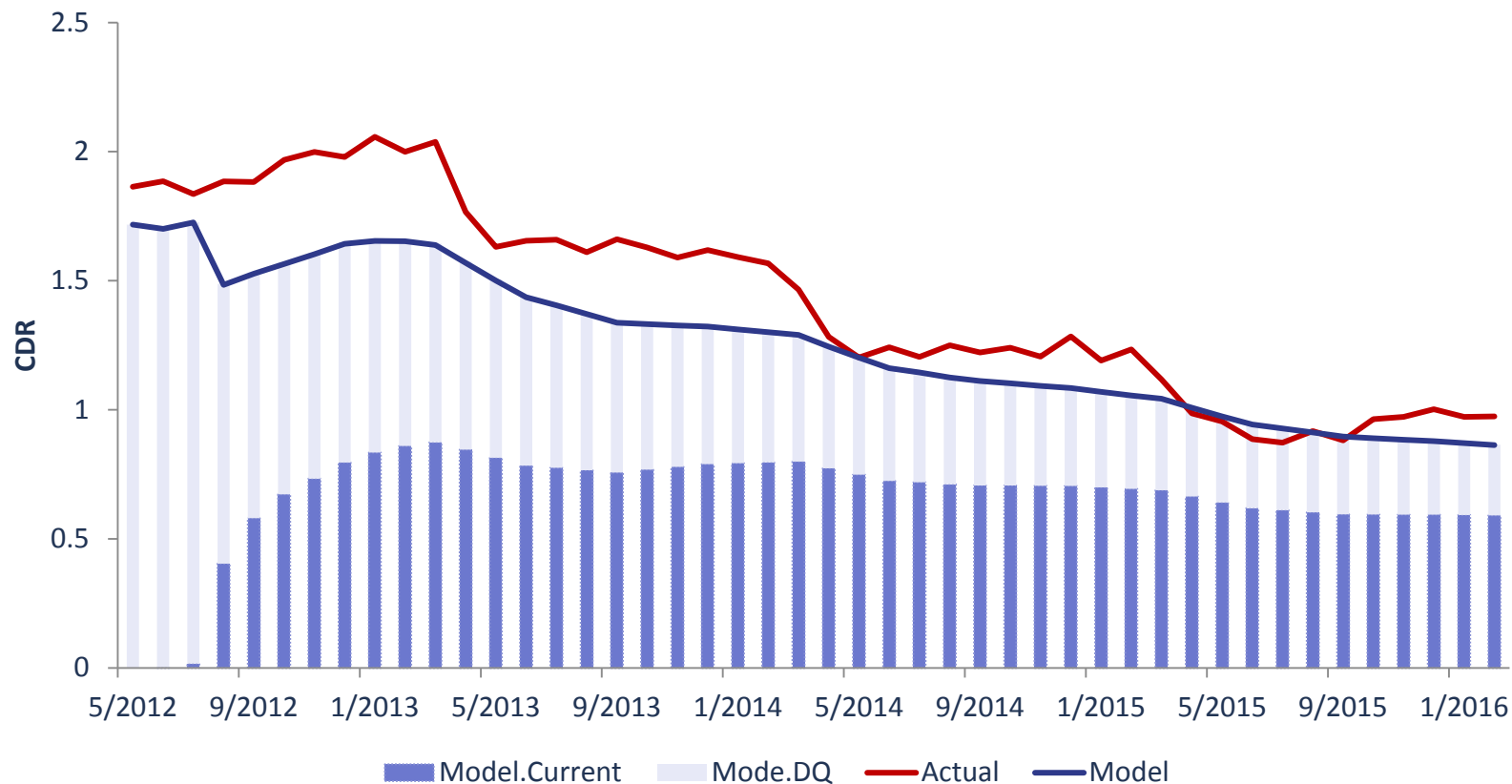


Model Tracking Performance – Historical



In order to examine model results for further back in history, we complement the pool-level data set with loan level default data (each loan is treated as a distinct pool by the pool-level model). Results show that even though we only use data from 2011 onwards to calibrate the default model, the v3 model modestly underperforms v1 for the out-of-sample historical period ranging from 1999 to 2011. And, for tracking from 2011 to 2016, v3 shows its strength.

Model Tracking Performance – Full Historical (For all existing pools as of 2012/05)



Based on a sample of all pools that exist as of 2012-05 (CQ, CR, CL etc.), we simulate the “out-of-sample” tracking power of the MonthsSinceLastPaymentStatus curve. The model’s forecasts for all months subsequent to 2012-05 are based on updated collateral/market conditions (HPA 2-year, WALA, CLTV) but only rely on the initial payment status of the relevant pools as of 2012-05.

01
Month Since Start

➤ Month since current / delinq status helps to predict the long term evolution of pool default rates

02
CLTV

➤ Borrower with higher CLTVs will have a higher risk of default.

03
Credit Availability

➤ When credit availability goes higher, borrowers will have a better chance to refinance lowers the default rate.

04
FICO

➤ Borrowers with lower FICO scores have higher risk of default.

05
WALA

➤ The WALA ramp effect shows a increasing trend of default as loan age goes up.

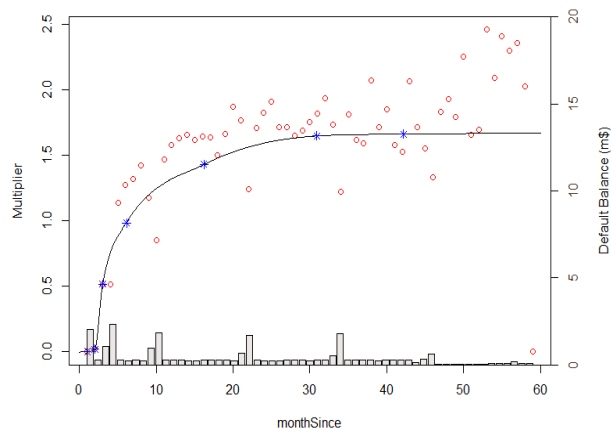
v3 of the default model includes the five prediction parameters listed on the left-hand side.

Sequentially fitting model with the five parameters, we can create a cubic spline curve for each one of these inputs.

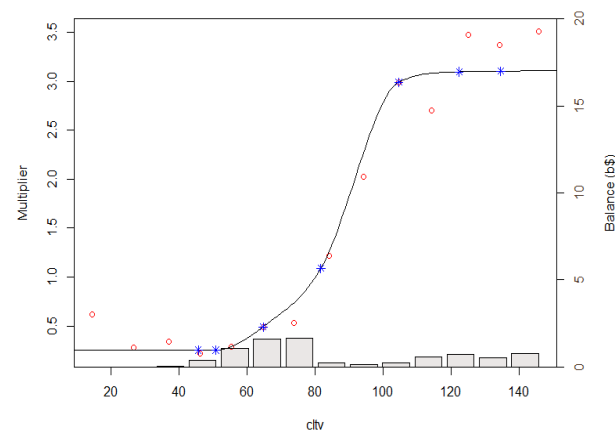
The parameter ordering is based on a single parameter regression calculated R-square value.

Parameters that have earlier order will have more significant impact on our model out put.

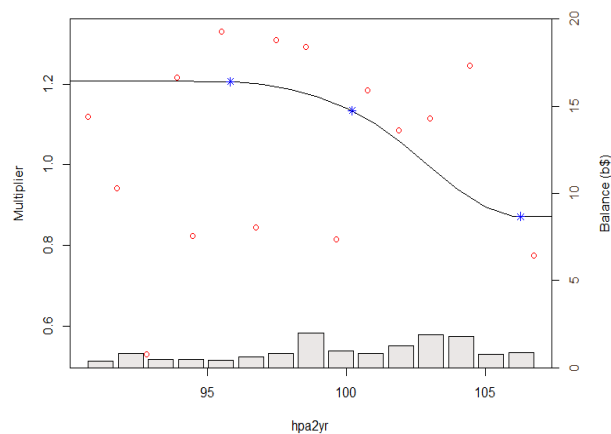
Spline Multiplier: monthSinceCurrent



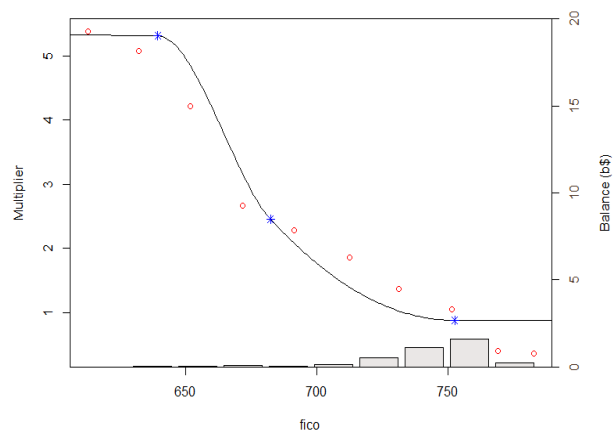
Spline Multiplier: citv



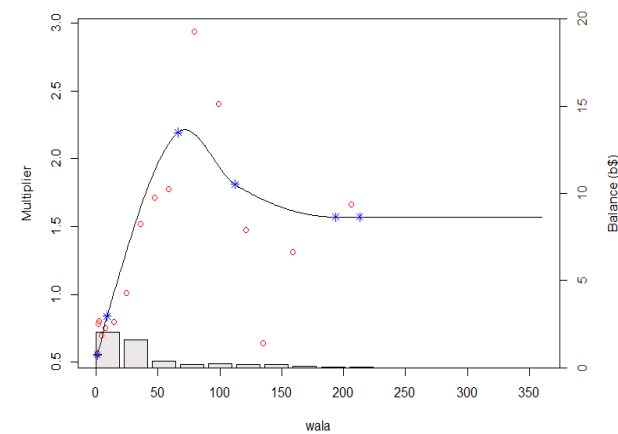
Spline Multiplier: hpa2yr



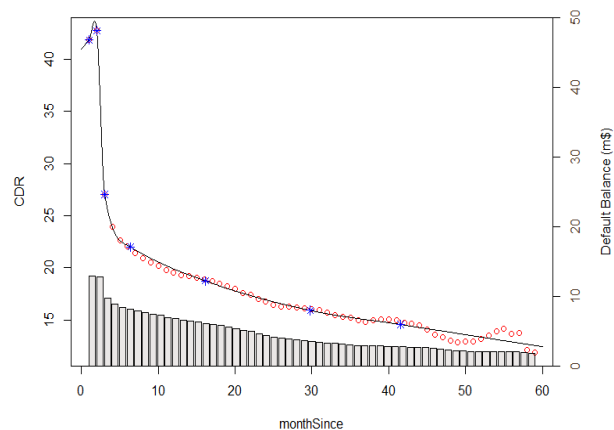
Spline Multiplier: fico



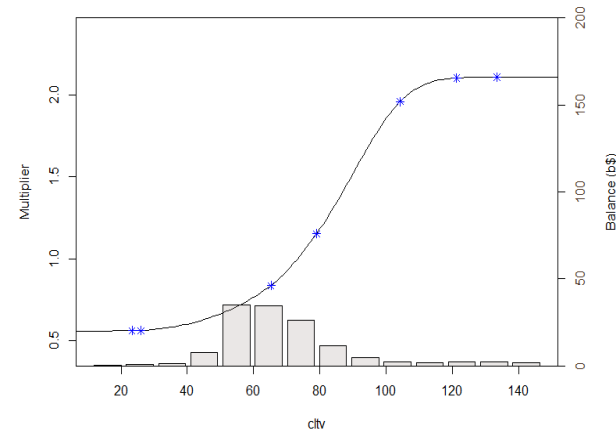
Spline Multiplier: wala



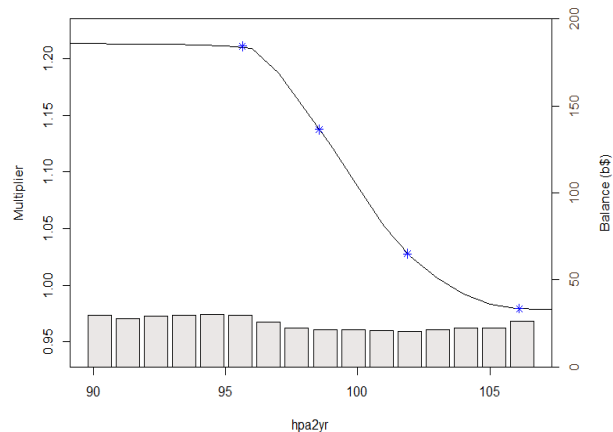
Spline Multiplier: monthSince



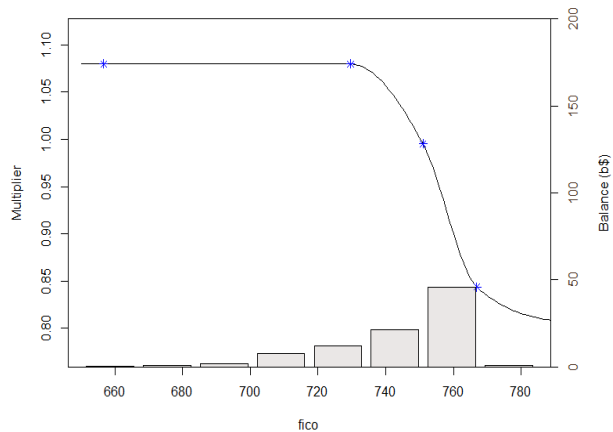
Spline Multiplier: cltv



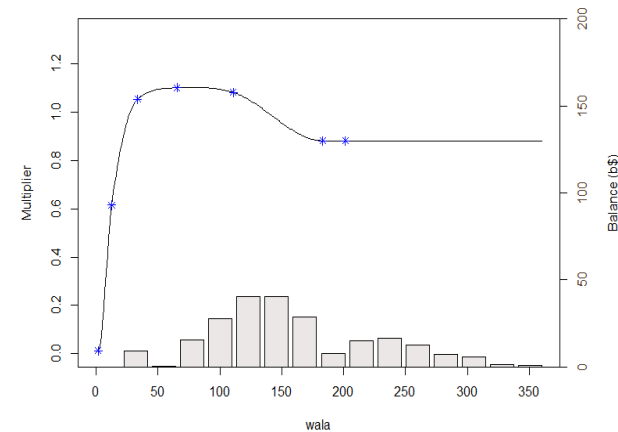
Spline Multiplier: hpa2yr



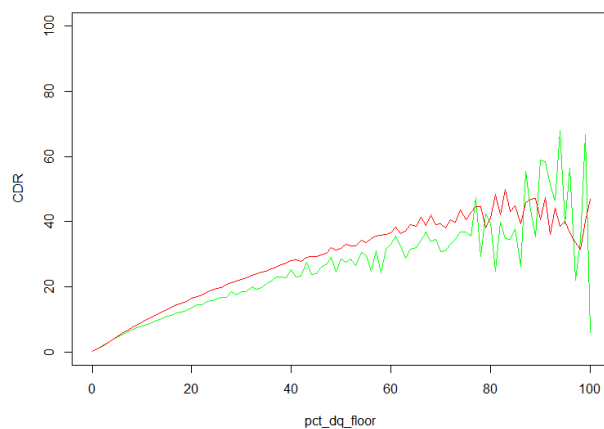
Spline Multiplier: fico



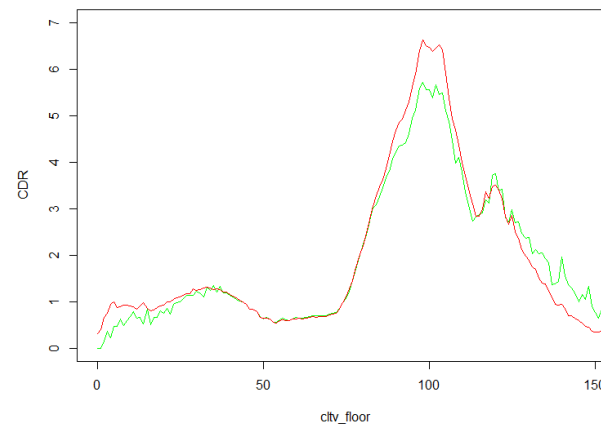
Spline Multiplier: wala



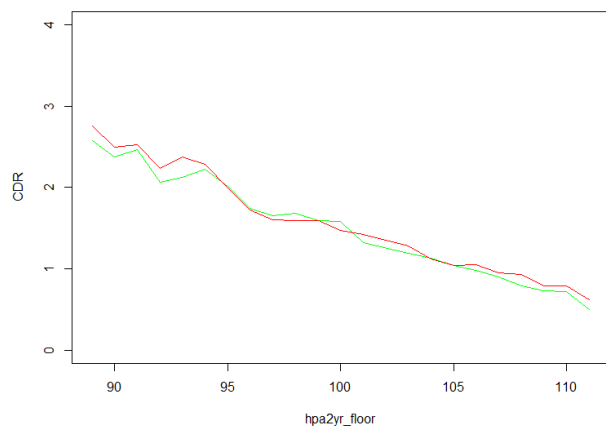
Delinq Percentage



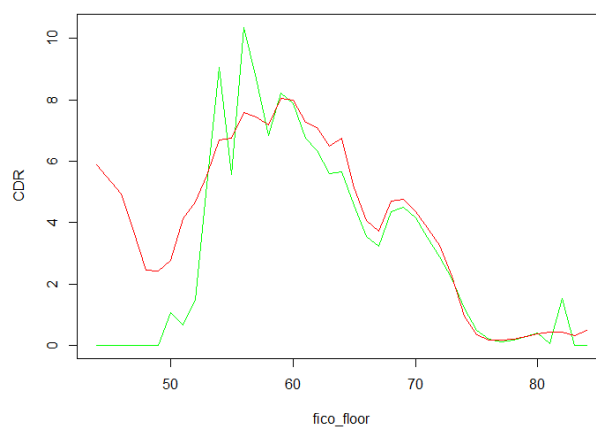
CLTV



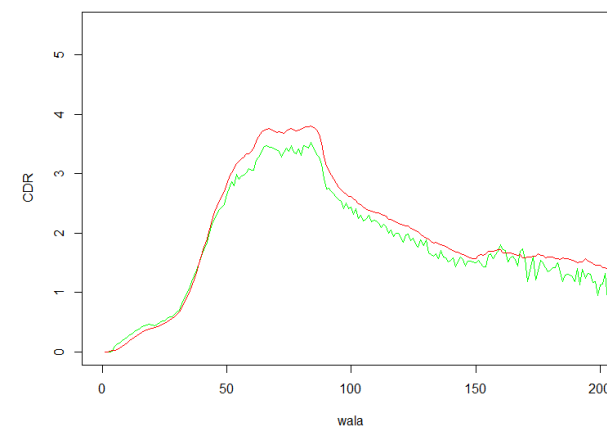
Credit Availability



FICO



WALA



Actual
Model

100% Delinquent Pools

	Min	Max
baselineConstant	44.29	44.29
cltv	0.56	2.11
hpa2yr	0.98	1.21
fico	0.80	1.08
wala	0.00	1.10
Default (%CPR)	0.000	84.8

Description of Default Data

- **Prepay Data**
 - Pool Level
 - Delinquency/Default information only available for Freddie pools
 - ~ 352 K pools with data as far back as 2011
 - 30 Day, 60 Day, 90 Day, 120 Day and Repurchased balances are available. These are exclusive, not cumulative percentages. The 120Day Balance is usually 0 since the Agencies will removed 120 Day delinquent loans from the pool but there are exceptions (see next slide)
 - The Delinq% is the sum of the 30Day, 60Day, 90Day and 120Day balances.
 - Loan Level
 - No Delinquency/Default information is available
- **Credit Data**
 - Delinquency/Default information available for Fannie and Freddie loans
 - ~ 40 M loans with data as far back as 1999
 - Loan Age up to 180
 - Has indicators for Prepayment, and Default
 - Excellent coverage for FICO, DTI, CLTV (msa level), Occupancy, etc.

Description of Default Data: Default Definition

- **Prepay Data**
 - Pool Level: Default Definition
 - 120 day delinquent loans are *usually* removed from the pool. From 2009-2011, the Agencies did not repurchase 120 day loans because they were suffering from funding issues.
 - Default balance is reported
- **Credit Data**
 - Loan Level: Default Definition
 - 180 day delinquent loans are removed from the data and a Zero Balance Code is reported
- **Revised Credit Data**
 - We must modify the default criteria to match with the Pool data
 - First time a loan reaches 120 days delinquent, the loan is marked as a default
- **Default Empirical Formula**
 - $\text{MDR (monthly default rate)} = \text{default balance} / \text{current balance}$

